SEBEI SCIENTISTS FORUM CONSTITUTION



July 8, 2022

1 Background

Sebei Scientists Forum (S.S.F) is an association that brings together all Scientists and Mathematicians in Sebei region. It includes but not limited to Agriculture, Biology, Chemistry, Economics, Engineering, Food Technology, Mathematics, Medicine, Physics, and any related courses.

Sebei Scientists Forum (S.S.F) formally called Sebei Association of Mathematical Scientists (known by the abbreviation S.A.M.S) was founded on the 5th May, 2020 by Ben Sorowen. The association is the first of its kind in sebei sub-region and its main focus is on Sebei Science, technology mathematics and the Education system in the region. Members agreed to change the name to Sebei Scientists Forum (S.S.F) to ensure inclusivity.

The members include University, College, High School and Primary Science teachers, Graduate, Undergraduate and College students undertaking Science, Technology, Engineering, and Mathematics (STEM) courses, Doctors, Pharmacists, Economists, Accountants, Agriculture, and many others in academia, Government, Business, and Industry and aims at bringing together all the Scientists and Mathematicians of Sebei region, improving the quality of teaching, identifying and nurturing talented people in science and mathematics, contributing to the development of science and mathematics and its applications to the science and technology in Sebei region.

We also welcome all who are interested in the Sciences and Technology.

2 Name, Address and Objective

2.1 Name

The Association shall be called "**Sebei Scientists Forum (S.S.F).**"

2.2 Address

The address of the association at any given time shall be the Principal place of employment of the President then in office.

This address shall be published in the associations Directory and News letter.

2.3 The Aims of the Association

The Aim of the association is **"To strive for the improvement of mathematics and Science education in Sebei region"**.

2.3.1 Objectives of the Association

The Objectives of the association are;

1. To engage in deep deliberation on pressing problems facing science and mathematics in Sebei region, and to propose appropriate recommendations to relevant authorities.

2. To promote the advancement of science and mathematics in Sebei region.
3. To facilitate the exchange of information about current research work and teaching methods between science and mathematics teachers in Sebei region.
4. To arrange Prize, or similar incentive for students who perform very well in regional mathematics competitions with the main goal of attracting students to mathematics in Sebei region.
5. To connect all the science and mathematics teachers in the Sebei region and organize common Examinations i.e end of term and end of year exams for both 'O' Level and 'A' Level.
6. To establish and maintain contact with similar associations outside Sebei region and exchange views.
7. To organize School outreaches and public lectures in Sebei region.
8. To promote interest in, and enjoyment of science and mathematics by members of Sebei region.
9. To work to educate all students and the community at large aimed at increasing the level of awareness of scientific issues, and providing a knowledge resource to the Sebei people.
10. To scale up the use of evidence based pedagogical methods in Sebei region.
11. Increasing the number of talented students who are interested in directing their studies towards science, technology and mathematics.

3 Mission

The mission for association shall be "**To enhance science, technology and mathematical excellence in Sebei region**"

4 Vision

The Vision for Association shall be;

- Inspire, empower and develop a new generation of scientists, engineers, medical personnel and mathematicians, dedicated to the highest principles of science and technology in Sebei region.
- A joint effort, focused on improving the quality of education and modernizing Education system in Sebei region.

5 Motto

The Motto for association shall be "**Achieving Excellence Together**"

6 Membership

6.1 Classes of members

The Association shall consist of the following classes of members:

(i) Honorary member

(ii) Ordinary member

(iii) Student member

6.2 Admission

(i) Honorary Member shall be someone who has made a significant contribution to the Association; or to the advancement of teaching and learning of a field in science and mathematics, or to its applications, or to the meaningful progress of science, technology, agriculture and mathematics teaching and their applications.

All mathematics teachers, Scientists, Engineers, Medical Personnel, Economists, all Graduates, Government and None Government Organization employees who apply mathematics, science and technology directly in their work places qualify to be under this category.

A Candidate for admission to Honorary Membership may be nominated at an Ordinary Meeting of the Society by General Committee only, following a proposal from at least three members of the Society, and shall be admitted if at least two thirds of the members present vote in favor.

(ii) Ordinary membership shall be open to all those who are interested in furthering the aims of the Association. Applications will be considered by the General Committee and shall be refused for good and proper reasons only.

Any person may apply for Ordinary membership.

(iii) A student in attendance at any institution of learning may apply for Student Membership.

(iv) Ordinary and Student Membership must be made in writing on an approved form obtained from the Honorary Secretary of the Association.

(v) All applications for membership shall be considered by the Council which shall have absolute discretion as to the approval of such applications.

6.3 Privileges of Membership

(i) Honorary, Ordinary and Student members shall be entitled to participate in the activities of the association, but the Ordinary and Student members shall have a right to attend and vote at all meetings of the association. Only Honorary and Ordinary members shall be entitled to hold office in the Executive Council.

(ii) Honorary, Ordinary and Student members shall be entitled to get a copy of each regular issue of the Bulletin published by the association

(iii) All members may obtain any publications of the association at special rates to be determined by the Council.

(iv) Non-members may be admitted to meetings of the association, provided that the President believes there is a good reason for attendance.

7 Subscription

(a) Honorary members shall not be required to pay any subscription.

(b) Each Ordinary member shall pay an annual subscription fee to be determine by the Council Members.

(c) Each student Member shall not be required to pay any subscription to the Association.

8 Membership Register

A register of members, with particulars of each member shall be maintained. Each member shall furnish the Sectary with an address for correspondence, and all material posted to such an address shall be considered to have been delivered.

9 Arrears and Resignation

An Ordinary member not having made a life composition and a student shall be liable for his/her annual subscription until he has given written notice of his resignation to the Secretary. A member whose subscription is in arrears shall not have the power to vote, and shall not be entitled to receive any material published by the society. The name of the Ordinary Member whose subscription is in arrears for two years and fails to pay it up within one month after a notice in writing has been served on him shall automatically be removed from the Membership Register.

10 Direction and Management of the Association

The direction and management of the affairs of the society shall be vested in the Council. The Council may, subject to the provisions of the Constitution, exercise all or any of the powers of the Council as provided by the Constitution.

10.1 Composition and Election of the Council

The Council shall consist of the

- President,
- Three Vice-Presidents,

- Secretary,
- Assistant Secretary,
- Treasurer,
- Committee Members.

The Council members shall be elected annually at an Annual General Meeting and shall hold office until the next Annual General Meeting. No Council member shall hold more than one office in the Council.

11 Powers of the Council

The powers of the Council are as follows:

(a) To carry out all activities of the Association in accordance with the aims of the association.

(b) To fill any vacancy in the Council or in the posts of Auditors and Returning Officer occurring between consecutive Annual General Meetings.

(c) To appoint and dissolve sub-committees and delegate to them such powers as it may determine. The President, Vice-Presidents and the Secretaries may sit on all such sub-committees as ex-officio members. Sub-committees shall automatically cease to exist on the expiry of the term of the Council which appoint them.

(d) To make a provisional decision upon any matter for which no provision is made in the Constitution of the association. Such decision shall be subject to ratification and/or revision at an Extraordinary General Meeting which shall be called not later than five weeks after such a decision is made.

(e) To accept any gift or donation with a view to furthering directly or indirectly the aims of the association.

(f) To borrow or raise any money that may be required upon such times as may be deemed necessary and in particular by mortgage or charge of all or any of the property of the association.

(g) To invest any monies in such manner from time to time as may determined.

(h) To purchase or lease, rent, hold or dispose of, construct or alter any equipment of building to be used as a College, library, offices of lecture rooms or any other property, real or personal, for the advancement of the aims of the association.

(i) To open an operate any class of bank account.

(j) To sell, manage, lease mortgage, dispose of or otherwise deal with all or part of the property of the association as may be deemed expedient with the view to the promotion of the aims of the association.

(k) To establish libraries and collections of equipment for the use of members and others and to publish any material that may be of interest to members in accordance with the aims of the association.

(l) To contribute to the expenses of persons engaged in authorized activities designed to promote the aims of the association.

(m) To appoint employees, to determine the conditions of their employment, and to suspend or remove them.

(n) To hold or participate in the holding of conferences and meetings, the reading of papers and the delivery of lectures, and to provide any other means for collecting and disseminating information concerning mathematics and its applications.

(o) To conduct or participate in the conduct of examinations for 'O'Level and 'A' Level in our region.

(p) To award prizes for outstanding academic or professional achievement to both Members and best performing students in regional mathematics competitions.

(q) To affiliate with any organization having aims compatible with those of the association.

12 Duties and Powers of Council Members

12.1 The President

The President shall preside over all meetings of the Association Council meetings. He shall represent the association either by himself or through his nominee from among the Council in the Associations dealings with outside persons and bodies. He shall also perform all other duties which are incumbent upon his office.

12.2 The Vice-Presidents

Either of the Vice-Presidents shall perform the duties of the President in the event of his absence. They shall also assist the President in carrying out his duties.

12.3 The Secretary

It shall be the duty of the Secretary to make and keep proper minutes of all meetings of the association and of the Council, to send out all notices required by the Constitution or as instructed by the Council and to do all things as is proper to his office.

12.4 The Assistant Secretary

It shall be the duty of the Assistant Secretary to assist the Secretary in carrying out his duties, and to act for him/her during his/her absence.

12.5 The Treasurer

It shall be the duty of the Treasurer to receive and pay all monies on behalf of the association and to cause proper records to be kept. All payments shall be made to him as directed by the Council. He shall prepare and submit an audited financial report to the Annual General Meeting. He shall keep a copy of the register of all members of the association. He shall have access to accounts of sub-committees appointed by the Council.

12.6 The Committee Members

The Committee Members of the Council shall assist the work of the other Council members in running the association, and be responsible for any such work of the association as may be delegated to them by the Council.

13 Returning Officer

The Returning Officer shall be elected annually at the Annual General Meeting and he shall be entirely responsible for conducting the elections of the incoming Council according to the provisions of the Constitution. He shall not be a member of the Council nor stand for election to the Council during the tenure of his appointment.

14 Meetings

Meetings of the Association shall be:

(a) **Ordinary General Meeting:** The time and place of every Ordinary General Meeting of the association shall be determined by the President, and announced before the date of the meeting.

(b) **Annual General Meeting:** The Annual General Meeting shall be held on a date, to be decided by the Council, after 1st June but not later than 31st August of each year and shall be announced by circular addressed to all members of the association not less than two weeks before the date of the meeting. The Members shall be notified at least five weeks prior to the date of the Annual General Meeting. Notices of the principal business to be transacted at the meeting shall be sent to all members with the circular announcing the meeting.

The main business of the Annual General Meeting shall be:

(i) To receive the Annual Report on the proceedings of the Association during the past year to be submitted and presented by the President.

(ii) To receive the audited financial report submitted by the Treasurer.

(iii) To consider any other matters of which notice in writing has been given to the Honorary Secretary at least seven days before the meeting.

(iv) To elect members of the Council.

(v) to access the progress of the Associations Aims.

(vi) To elect the Returning Officer.

(c) **Extraordinary General Meeting:** An Extraordinary General Meeting shall be held if deemed expedient by the Council. In addition, the President shall call such a meeting at the written request of twelve members who shall specify the motions which they wish to propose at the meeting. Such meeting shall be called within four weeks of receiving the request. The business at such a meeting must be confined to the specified motions. At least seven days notice of such a meeting shall be given to all members

(d) **Council Meeting:** The Council shall meet at such times and places as may be decided from time to time by the Council or deemed necessary by the President, but shall meet at least three times a year. At least two days notice of any meeting shall be given to Council members.

15 Quorum

(a) **Annual General Meeting and Extraordinary General Meeting**. At any such Meeting, twenty members or one-third of the total membership shall form a quorum. In the event of a quorum not being obtained, the meeting shall be adjourned to a date not earlier than ten days and not later than twenty days from the date of the meeting. Notice of the adjourned meeting shall be given to members at least seven days before the meeting. If at the adjourned meeting there is no quorum the meeting shall proceed as though the quorum has been obtained.

(b) **Council Meeting** At any Council Meeting, five Council members shall form a quorum. Voting at such Council meeting shall be by show of hands if agreed to by all Council members present, otherwise it shall be by secret ballot. Each member shall have one vote. All resolutions and decisions not governed shall be carried by a simple majority vote. In the event of equal votes, the President shall have a casting vote.

16 Conduct of Business

A Council member who is absent from three consecutive Council Meeting without leave may, following a resolution of the Council, be asked to show good cause why his position on the Council should not be declared vacant. At its discretion, the Council may then proceed to declare his position vacant.

17 Election

(a) Not less than five weeks prior to the date set for the Annual General Meeting, the Returning Officer shall notify all members of the election and shall call for nomination for election to the Council.

(b) Persons nominated for election must not be in arrears with their subscriptions. All nominations shall be signed by two members and lodged with the Returning Officer not less than two weeks prior to the date set for the Annual General Meeting. Each nomination shall state the position to which election is sought and be signed by the candidate as evidence of his willingness to

accept nomination. A person may not be nominated for more than one position as Office bearer.

(c) If the Returning Officer receives more nominations for any position on the Council than there are vacancies he shall conduct a ballot among the members at the Annual General Meeting. Such a ballot shall be determined by a simple majority of the votes cast and the results shall be declared at the Annual General Meeting.

(d) If the number of nominations received by the Returning Officer for any position on the Council is equal to the number of vacancies, the Returning Officer shall at the Annual General Meeting declare such nominees duly elected to the Council.

(e) If the number of nominations received by the Returning Officer for any position on the Council is less than the number of vacancies, then at the Annual General Meeting he shall in respect of those positions

(i) declare elected those members who have been duly nominated,

(ii) call for further nominations for the positions to be filled and,

(iii) where necessary, conduct ballots amongst the members to determine by simple majority which of the further candidates should fill the vacancies. The results of such ballots shall be declared before the close of the Annual General Meeting.

(f) In the case of equal votes the outgoing President shall have the casting vote.

18 Amendment of the Constitution

(a) This Constitution shall be revoked, altered or amended only by resolution at the Annual General Meeting or Extraordinary General Meeting by a majority of not less than two third of the members present and voting. All proposed revocation, alteration, or amendments shall be made in writing to the Honorary Secretary. All amendments to the Constitution shall only be enforced after sanction by the Registrar of associations.

(b) All members shall be informed of any proposed amendment to the Constitution at least two weeks before the date of the meeting at which the proposal is considered.

19 Property

(a) The property of the association shall be the monies received by way of any subscription, fees and gifts or donations made to the association and such other property into which these monies may be subsequently converted.

(b) The income and property of the association, whence so ever derived shall be applied solely towards the promotion of the aims of the association and no portion thereof shall be paid or transferred directly or indirectly by way of dividends, bonus or otherwise howsoever by way of profit, to the members of the association provided that nothing herein shall prevent the payment in good faith, of reasonable and proper remuneration to any employee of the association, or to any member of the association in return for any service actually rendered to the association.

20 Prohibitions

(1) A member of the association must not;

(i) Engage in any trade union activities.

(ii) Hold any lottery, whether confined to its members or not, in the name of the association.

(iii) Indulge in any political activity or allow association funds and/or premises to be used for political purposes.

(iv) Carry out parallel activities that can sabotage the operations of the association.

(v) Use information obtained from the register of members of the association to contact, or send material to, or another member of the association for the purpose of advertising for political, charitable or commercial purposes; or

(vi) Disclose information obtained from the register to someone else, knowing that the information is likely to be used to contact, or send material to, another member of the association for the purpose of advertising for political, charitable or commercial purposes.

(2) Sub-rule (1) does not apply if the use or disclosure of the information is approved by the association.

21 Rules

A few of the association's rules

1. The Association shall be called the Sebei Scientists Forum (S.S.F).

2. The object of the Association shall be the promotion of activities pertaining to the sciences, technology and mathematics in sebei region.

3. All science professionals of sebei region, and no others, may apply for Ordinary Membership. Undergraduate, College and secondary students, and no others, may apply for Student membership. Senior citizens of sebei region and in special cases those who have made great contribution to community and others may be invited to Honorary Membership.

4. The affairs of the Association shall be managed by a Committees consisting of at least four Officers, who shall be Ordinary Members of the Society: a President, a Vice-President, a Secretary, a Treasurer, and up to three others. The roles of the Committees shall be detailed in the Standing Orders. The partial map from Officers to Ordinary Members shall be injective.

22 Science, Technology, Engineering and Mathematics Careers / Occupations

22.1 Geological and Petroleum Technician

Assist scientists or engineers in the use of electronic, sonic, or nuclear measuring instruments in both laboratory and production activities to obtain data indicating potential resources such as metallic ore, minerals, gas, coal, or petroleum. Analyze mud and drill cuttings. Chart pressure, temperature, and other characteristics of wells or bore holes. Investigate and collect information leading to the possible discovery of new metallic ore, minerals, gas, coal, or petroleum deposits.

22.2 Geophysical Data Technician

Measure, record, or evaluate geological data, using sonic, electronic, electrical, seismic, or gravity-measuring instruments to prospect for oil or gas. May collect or evaluate core samples or cuttings.

22.3 Forensic Science Technician

Collect, identify, classify, and analyze physical evidence related to criminal investigations. Perform tests on weapons or substances, such as fiber, hair, and tissue to determine significance to investigation. May testify as expert witnesses on evidence or crime laboratory techniques. May serve as specialists in area of expertise, such as ballistics, fingerprinting, handwriting, or biochemistry

22.4 Architectural and Engineering Manager

Plan, direct, or coordinate activities in such fields as architecture and engineering or research and development in these fields.

22.5 Biofuels/Biodiesel Technology and Product Development Manager

Define, plan, or execute biofuels/biodiesel research programs that evaluate alternative feedstock and process technologies with near-term commercial potential.

22.6 Mathematician

Conduct research in fundamental mathematics or in application of mathematical techniques to science, management, and other fields. Solve problems in various fields using mathematical methods.

22.7 Statistician

Develop or apply mathematical or statistical theory and methods to collect, organize, interpret, and summarize numerical data to provide usable information. May specialize in fields such as biostatistics, agricultural statistics, business statistics, or economic statistics. Includes mathematical and survey statisticians.

22.8 Biostatistician

Develop and apply biostatistical theory and methods to the study of life sciences.

22.9 Clinical Data Manager

Apply knowledge of health care and database management to analyze clinical data, and to identify and report trends.

22.10 Mathematical Technician

Apply standardized mathematical formulas, principles, and methodology to technological problems in engineering and physical sciences in relation to specific industrial and research objectives, processes, equipment, and products.

22.11 Cartographer and Photogrammetrist

Collect, analyze, and interpret geographic information provided by geodetic surveys, aerial photographs, and satellite data. Research, study, and prepare maps and other spatial data in digital or graphic form for legal, social, political, educational, and design purposes. May work with Geographic Information Systems (GIS). May design and evaluate algorithms, data structures, and user interfaces for GIS and mapping systems

22.12 Aerospace Engineer

Perform engineering duties in designing, constructing, and testing aircraft, missiles, and spacecraft. May conduct basic and applied research to evaluate adaptability of materials and equipment to aircraft design and manufacture. May recommend improvements in testing equipment and techniques.

22.13 Computer Hardware Engineer

Research, design, develop, or test computer or computer-related equipment for commercial, industrial, military, or scientific use. May supervise the manufacturing and installation of computer or computer-related equipment and components.

22.14 Electrical Engineer

Research, design, develop, test, or supervise the manufacturing and installation of electrical equipment, components, or systems for commercial, industrial, military, or scientific use.

22.15 Electronics Engineers, Except Computer

Research, design, develop, or test electronic components and systems for commercial, industrial, military, or scientific use employing knowledge of electronic theory and materials properties. Design electronic circuits and components for use in fields such as telecommunications, aerospace guidance and propulsion control, acoustics, or instruments and controls.

22.16 Radio Frequency Identification Device Specialist

Design and implement radio frequency identification device (RFID) systems used to track shipments or goods

22.17 Health and Safety Engineer, Except Mining Safety Engineer and Inspector

Promote worksite or product safety by applying knowledge of industrial processes, mechanics, chemistry, psychology, and industrial health and safety laws. Includes industrial product safety engineers.

22.18 Industrial Safety and Health Engineer

Plan, implement, and coordinate safety programs, requiring application of engineering principles and technology, to prevent or correct unsafe environmental working conditions.

22.19 Fire-Prevention and Protection Engineer

Research causes of fires, determine fire protection methods, and design or recommend materials or equipment such as structural components or fire-detection equipment to assist organizations in safeguarding life and property against fire, explosion, and related hazards.

22.20 Product Safety Engineer

Develop and conduct tests to evaluate product safety levels and recommend measures to reduce or eliminate hazards.

22.21 Industrial Engineer

Design, develop, test, and evaluate integrated systems for managing industrial production processes, including human work factors, quality control, inventory control, logistics and material flow, cost analysis, and production coordination.

22.22 Human Factors Engineer and Ergonomist

Design objects, facilities, and environments to optimize human well-being and overall system performance, applying theory, principles, and data regarding the relationship between humans and respective technology. Investigate and analyze characteristics of human behavior and performance as it relates to the use of technology.

22.23 Marine Engineers and Naval Architect

Design, develop, and evaluate the operation of marine vessels, ship machinery, and related equipment, such as power supply and propulsion systems.

22.24 Marine Engineer

Design, develop, and take responsibility for the installation of ship machinery and related equipment including propulsion machines and power supply systems.

22.25 Marine Architect

Design and oversee construction and repair of marine craft and floating structures such as ships, barges, tugs, dredges, submarines, torpedoes, floats, and buoys. May confer with marine engineers.

22.26 Materials Engineer

Evaluate materials and develop machinery and processes to manufacture materials for use in products that must meet specialized design and performance specifications. Develop new uses for known materials. Includes those engineers working with composite materials or specializing in one type of material, such as graphite, metal and metal alloys, ceramics and glass, plastics and polymers, and naturally occurring materials. Includes metallurgists and metallurgical engineers, ceramic engineers, and welding engineers.

22.27 Mechanical Engineer

Perform engineering duties in planning and designing tools, engines, machines, and other mechanically functioning equipment. Oversee installation, operation, maintenance, and repair of equipment such as centralized heat, gas, water, and steam systems.

22.28 Fuel Cell Engineer

Design, evaluate, modify, or construct fuel cell components or systems for transportation, stationary, or portable applications.

22.29 Automotive Engineer

Develop new or improved designs for vehicle structural members, engines, transmissions, or other vehicle systems, using computer-assisted design technology. Direct building, modification, or testing of vehicle or components.

22.30 Mining and Geological Engineer, Including Mining Safety Engineer

Conduct sub-surface surveys to identify the characteristics of potential land or mining development sites. May specify the ground support systems, processes and equipment for safe, economical, and environmentally sound extraction or underground construction activities. May inspect areas for unsafe geological conditions, equipment, and working conditions. May design, implement, and coordinate mine safety programs.

22.31 Nuclear Engineer

Conduct research on nuclear engineering projects or apply principles and theory of nuclear science to problems concerned with release, control, and use of nuclear energy and nuclear waste disposal.

22.32 Petroleum Engineer

Devise methods to improve oil and gas extraction and production and determine the need for new or modified tool designs. Oversee drilling and offer technical advice.

22.33 Biochemical Engineer

Develop usable, tangible products, using knowledge of biology, chemistry, or engineering. Solve problems related to materials, systems, or processes that interact with humans, plants, animals, microorganisms, or biological materials.

22.34 Validation Engineer

Design or plan protocols for equipment or processes to produce products meeting internal and external purity, safety, and quality requirements.

22.35 Energy Engineer

Design, develop, or evaluate energy-related projects or programs to reduce energy costs or improve energy efficiency during the designing, building, or remodeling stages of construction. May specialize in electrical systems; heating, ventilation, and air-conditioning (HVAC) systems; green buildings; lighting; air quality; or energy procurement.

22.36 Manufacturing Engineer

Design, integrate, or improve manufacturing systems or related processes. May work with commercial or industrial designers to refine product designs to increase producibility and decrease costs.

22.37 Mechatronics Engineer

Research, design, develop, or test automation, intelligent systems, smart devices, or industrial systems control.

22.38 Microsystems Engineer

Research, design, develop, or test microelectromechanical systems (MEMS) devices.

22.39 Photonics Engineer

Design technologies specializing in light information or light energy, such as laser or fiber optics technology.

22.40 Robotics Engineer

Research, design, develop, or test robotic applications.

22.41 Nanosystems Engineer

Design, develop, or supervise the production of materials, devices, or systems of unique molecular or macromolecular composition, applying principles of nanoscale physics and electrical, chemical, or biological engineering.

22.42 Wind Energy Engineer

Design underground or overhead wind farm collector systems and prepare and develop site specifications.

22.43 Solar Energy Systems Engineer

Perform site-specific engineering analysis or evaluation of energy efficiency and solar projects involving residential, commercial, or industrial customers. Design solar domestic hot water and space heating systems for new and existing structures, applying knowledge of structural energy requirements, local climates, solar technology, and thermodynamics.

22.44 Biologist

Research or study basic principles of plant and animal life, such as origin, relationship, development, anatomy, and functions.

22.45 Biochemists and Biophysicist

Study the chemical composition or physical principles of living cells and organisms, their electrical and mechanical energy, and related phenomena. May conduct research to further understanding of the complex chemical combinations and reactions involved in metabolism, reproduction, growth, and heredity. May determine the effects of foods, drugs, serums, hormones, and other substances on tissues and vital processes of living organisms.

22.46 Zoologists and Wildlife Biologist

Study the origins, behavior, diseases, genetics, and life processes of animals and wildlife. May specialize in wildlife research and management. May collect and analyze biological data to determine the environmental effects of present and potential use of land and water habitats.

22.47 Bioinformatics Scientist

Conduct research using bioinformatics theory and methods in areas such as pharmaceuticals, medical technology, biotechnology, computational biology, proteomics, computer information science, biology and medical informatics. May design databases and develop algorithms for processing and analyzing genomic information, or other biological information.

22.48 Molecular and Cellular Biologist

Research and study cellular molecules and organelles to understand cell function and organization.

22.49 Geneticist

Research and study the inheritance of traits at the molecular, organism or population level. May evaluate or treat patients with genetic disorders.

22.50 Conservation Scientist

Manage, improve, and protect natural resources to maximize their use without damaging the environment. May conduct soil surveys and develop plans to eliminate soil erosion or to protect rangelands. May instruct farmers, agricultural production managers, or ranchers in best ways to use crop rotation, contour plowing, or terracing to conserve soil and water; in the number and kind of livestock and forage plants best suited to particular ranges; and in range and farm improvements, such as fencing and reservoirs for stock watering.

22.51 Soil and Water Conservationist

Plan or develop coordinated practices for soil erosion control, soil or water conservation, or sound land use.

22.52 Range Manager

Research or study range land management practices to provide sustained production of forage, livestock, and wildlife.

22.53 Medical Scientists, Except Epidemiologist

Conduct research dealing with the understanding of human diseases and the improvement of human health. Engage in clinical investigation, research and development, or other related activities. Includes physicians, dentists, public health specialists, pharmacologists, and medical pathologists who primarily conduct research.

22.54 Astronomer

Observe, research, and interpret astronomical phenomena to increase basic knowledge or apply such information to practical problems.

22.55 Physicist

Conduct research into physical phenomena, develop theories on the basis of observation and experiments, and devise methods to apply physical laws and theories.

22.56 Atmospheric and Space Scientist

Investigate atmospheric phenomena and interpret meteorological data, gathered by surface and air stations, satellites, and radar to prepare reports and forecasts for public and other uses. Includes weather analysts and forecasters whose functions require the detailed knowledge of meteorology.

22.57 Chemist

Conduct qualitative and quantitative chemical analyses or experiments in laboratories for quality or process control or to develop new products or knowledge.

22.58 Materials Scientist

Research and study the structures and chemical properties of various natural and synthetic or composite materials, including metals, alloys, rubber, ceramics, semiconductors, polymers, and glass. Determine ways to strengthen or combine materials or develop new materials with new or specific properties for use in a variety of products and applications. Includes glass scientists, ceramic scientists, metallurgical scientists, and polymer scientists.

22.59 Environmental Scientists and Specialist, Including Health

Conduct research or perform investigation for the purpose of identifying, abating, or eliminating sources of pollutants or hazards that affect either the environment or the health of the population. Using knowledge of various scientific disciplines, may collect, synthesize, study, report, and recommend action based on data derived from measurements or observations of air, food, soil, water, and other sources.

22.60 Climate Change Analyst

Research and analyze policy developments related to climate change. Make climate-related recommendations for actions such as legislation, awareness campaigns, or fundraising approaches.

22.61 Environmental Restoration Planner

Collaborate with field and biology staff to oversee the implementation of restoration projects and to develop new products. Process and synthesize complex scientific data into practical strategies for restoration, monitoring or management.

22.62 Industrial Ecologist

Apply principles and processes of natural ecosystems to develop models for efficient industrial systems. Use knowledge from the physical and social sciences to maximize effective use of natural resources in the production and use of goods and services. Examine societal issues and their relationship with both technical systems and the environment.

22.63 Geoscientist, Except Hydrologist and Geographer

Study the composition, structure, and other physical aspects of the Earth. May use geological, physics, and mathematics knowledge in exploration for oil, gas, minerals, or underground wa-

ter; or in waste disposal, land reclamation, or other environmental problems. May study the Earth's internal composition, atmospheres, oceans, and its magnetic, electrical, and gravitational forces. Includes mineralogists, crystallographers, paleontologists, stratigraphers, geodesists, and seismologists.

22.64 Hydrologist

Research the distribution, circulation, and physical properties of underground and surface waters; and study the form and intensity of precipitation, its rate of infiltration into the soil, movement through the earth, and its return to the ocean and atmosphere. beginsubsectionRemote Sensing Scientist and Technologist Apply remote sensing principles and methods to analyze data and solve problems in areas such as natural resource management, urban planning, or homeland security. May develop new sensor systems, analytical techniques, or new applications for existing systems.

22.65 Economist

Conduct research, prepare reports, or formulate plans to address economic problems related to the production and distribution of goods and services or monetary and fiscal policy. May collect and process economic and statistical data using sampling techniques and econometric methods.

22.66 Environmental Economist

Conduct economic analysis related to environmental protection and use of the natural environment, such as water, air, land, and renewable energy resources. Evaluate and quantify benefits, costs, incentives, and impacts of alternative options using economic principles and statistical techniques.

22.67 Neuropsychologist and Clinical Neuropsychologist

Apply theories and principles of neuropsychology to diagnose and treat disorders of higher cerebral functioning.

22.68 Pharmacists

Pharmacists are medication experts and play a critical role in helping people get the best results from their medications. Pharmacists prepare and dispense prescriptions, ensure medicines and doses are correct, prevent harmful drug interactions, and counsel patients on the safe and appropriate use of their medications.

They have specialized expertise about the composition of medicines, including their chemical, biological, and physical properties, as well as their manufacture and use. Other health care professionals rely on pharmacists to select and administer medications that offer the best results and quality of life for a particular patient.

Pharmacists may also prepare personalized medications, participate in patient rounds at a hospital, reduce the spread of infections, conduct research or clinical trials, or focus on a specific patient

population or disease state (e.g., diabetes, heart disease, asthma, HIV, and pain management).

22.69 Radiographer

Radiographers, also called radiologic technologists, are health care professionals who operate special scanning machines that make images for medical purposes. They use equipment like X-ray machines, CT scanners, and advanced technologies such as digital fluoroscopy. Radiographers often pass their findings on to radiologists, who interpret the images to help make a diagnosis. This is called diagnostic radiology. Interventional radiology uses imaging during a medical procedure to guide and assist in treatment. Besides a medical centerâs radiology department, radiologic technologists can work in areas like surgery, the emergency room, cardiac care, intensive care, and patient rooms.

22.70 Urban and Regional Planner

Develop comprehensive plans and programs for use of land and physical facilities of jurisdictions, such as towns, cities, counties, and metropolitan areas.

22.71 Anthropologist and Archeologist

Study the origin, development, and behavior of human beings. May study the way of life, language, or physical characteristics of people in various parts of the world. May engage in systematic recovery and examination of material evidence, such as tools or pottery remaining from past human cultures, in order to determine the history, customs, and living habits of earlier civilizations. beginsubsectionAnthropologist Research, evaluate, and establish public policy concerning the origins of humans; their physical, social, linguistic, and cultural development; and their behavior, as well as the cultures, organizations, and institutions they have created.

22.72 Archeologist

Conduct research to reconstruct record of past human life and culture from human remains, artifacts, architectural features, and structures recovered through excavation, underwater recovery, or other means of discovery.

22.73 Geographer

Study the nature and use of areas of the Earth's surface, relating and interpreting interactions of physical and cultural phenomena. Conduct research on physical aspects of a region, including land forms, climates, soils, plants, and animals, and conduct research on the spatial implications of human activities within a given area, including social characteristics, economic activities, and political organization, as well as researching interdependence between regions at scales ranging from local to global.

22.74 Forest and Conservation Technician

Provide technical assistance regarding the conservation of soil, water, forests, or related natural resources. May compile data pertaining to size, content, condition, and other characteristics of forest tracts, under the direction of foresters; or train and lead forest workers in forest propagation, fire prevention and suppression. May assist conservation scientists in managing, improving, and protecting rangelands and wildlife habitats.

22.75 Quality Control Analyst

Conduct tests to determine quality of raw materials, bulk intermediate and finished products. May conduct stability sample tests.

22.76 Precision Agriculture Technician

Apply geospatial technologies, including geographic information systems (GIS) and Global Positioning System (GPS), to agricultural production or management activities, such as pest scouting, site-specific pesticide application, yield mapping, or variable-rate irrigation. May use computers to develop or analyze maps or remote sensing images to compare physical topography with data on soils, fertilizer, pests, or weather.

22.77 Data Scientist

Develop and implement a set of techniques or analytics applications to transform raw data into meaningful information using data-oriented programming languages and visualization software. Apply data mining, data modeling, natural language processing, and machine learning to extract and analyze information from large structured and unstructured datasets. Visualize, interpret, and report data findings. May create dynamic data reports

22.78 Communications engineer

Communications engineers work within the broader field of electronic engineering and apply their knowledge of engineering principles to design solutions and advise management and other personnel on technical issues. They typically study, develop, design and maintain electronic systems and components for scientific, military, industrial or commercial use.

22.79 Agricultural engineer

Agricultural engineers are specialists who apply their knowledge of biology and various engineering disciplines â mainly chemical, civil, electrical and mechanical engineering â to design machinery that is used in agricultural production and processing.

22.80 Meteorologist

A meteorologist is an individual with specialized education who uses scientific principles to explain, understand,observe or forecast the earth's atmospheric phenomena and/or how the atmosphere affects the earth and life on the planet. Activities of a meteorologist often are classified into a number of specialized areas. A few examples are; air pollution meteorologist, global climate modeling, hydrometeorology, and numerical analysis and forecasting.

22.81 Food engineer

A food engineer is a type of engineer who works only with food products. Food engineers use the latest innovative products and technologies to process, preserve, and package and improve food products.

22.82 Systems engineer

A systems engineer's primary focus is how to design or manage an engineering project. I.e. clientsâ needs during the system building process, from project definition to design implementation.

22.83 Broadcast engineer

Broadcast engineers work in a subfield of electrical engineering that deals with radio and television broadcasting.

22.84 Chemical engineer

Chemical engineers apply their knowledge in chemical engineering to turn raw materials and chemicals into a wide range of new products. Their main task is to develop new, more economic ways of using materials, chemical substances, and energy.

22.85 Industrial Chemist

Industrial chemists assist chemical engineers in transforming small-scale chemical discoveries into mass-produced goods by analyzing compounds and substances to find their precise chemical or physical properties. Industrial chemists may also develop or improve products, formulas, analytical methods, and processes.

22.86 Biomedical engineer

Biomedical engineers use the principles of design and engineering and they operate in the field of biology and medicine. They typically research, design, or modify medical equipment and products.

22.87 Automotive engineer

Automotive engineers work in a branch of vehicle engineering that deals with the design, production and operation of automobiles, trucks, buses, motorcycles, and similar ground-based vehicles.

22.88 Power engineer

Power engineers, or power systems engineers, work within the broader field of energy engineering and their job primarily involves designing and developing renewable, energy efficent power grids and utility systems. They research and implement ways to integrate renewable power technology and are often involved in public works projects, and employed on federal projects that involve the integration of large power grids.


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